Task3_One_Page_Summary - Soteria SOW Guardian

Option B - Architecture Design | Entity: OCBC Bank Ltd | Jurisdictions: Singapore + United States | Domain: AI governance for AML/KYC source-of-wealth review

Soteria SOW Guardian is a governance layer for banks using AI in KYC source-of-wealth review. It is designed around OCBC Bank Ltd and Bank of Singapore. Bank of Singapore has announced a Source of Wealth Assistant, described as an agentic AI tool that can reduce source-of-wealth report preparation from around 10 days to one hour while relationship managers still review and refine the AI drafts before internal AML/CFT review [R2]. OCBC also has US agencies in New York and Los Angeles, making a Singapore-US comparison operationally relevant [R1].

The tool does not replace the bank's AI drafting assistant. It sits around the AI-assisted workflow and asks: What jurisdiction is active? What type of AI is being used? Are the AI-generated wealth claims supported by evidence? Did a human reviewer meaningfully challenge the output? What audit record shows that the bank governed the workflow properly?

The core design issue is that AI can make weak evidence look strong. The tool calculates Evidence Coverage Ratio, Unsupported Claim Rate, Contradiction Count, Human Challenge Rate, Override Quality Score, and Model Drift Flag. If evidence is below threshold, unsupported claims are above threshold, or material contradictions remain unresolved, the case enters Red Friction and cannot proceed to straight-through approval.

The jurisdiction logic is real rather than decorative. In Singapore mode, the tool activates MAS AI Model Risk Management, FEAT/Veritas-style responsible AI checks, and MAS Notice 626 AML/KYC controls [R3-R6]. In US mode, it activates FinCEN CDD controls and distinguishes traditional/non-generative models from generative or agentic AI [R7-R9]. For US generative/agentic AI cases, it issues a governance gap warning rather than falsely marking the system as fully covered by the 2026 model-risk guidance.

The same synthetic case produces different outputs. A Singapore case using agentic AI triggers MAS AI governance and AML/KYC source-of-wealth controls. A US case using the same agentic AI triggers FinCEN CDD plus a governance gap warning. A cross-border case activates dual-rule review and a no-silent-downgrade rule: a governance gap is not treated as permission to reduce controls.

The tool outputs Green, Amber, Red Friction, Amber + Governance Gap, and Red Friction + Governance Gap. It also produces an audit pack showing active rule versions, evidence maps, unresolved contradictions, reviewer actions, model drift alerts, and privacy exceptions. This solves a problem that a memo or spreadsheet cannot: continuous claim-to-evidence checking, rule versioning, drift monitoring, human challenge monitoring, and jurisdiction-specific control selection.

The tool does not approve customers automatically, file STRs/SARs automatically, certify that a case is 'MAS compliant' or 'US compliant,' replace relationship managers or compliance officers, or prove that wealth is legitimate. Its narrower purpose is to show whether the bank's conclusion is supported by available evidence, whether humans exercised judgment, and whether the correct jurisdictional governance logic was applied.

References: [R1]-[R9] are listed with full URLs in sources/references.md and in the Architecture Design document.